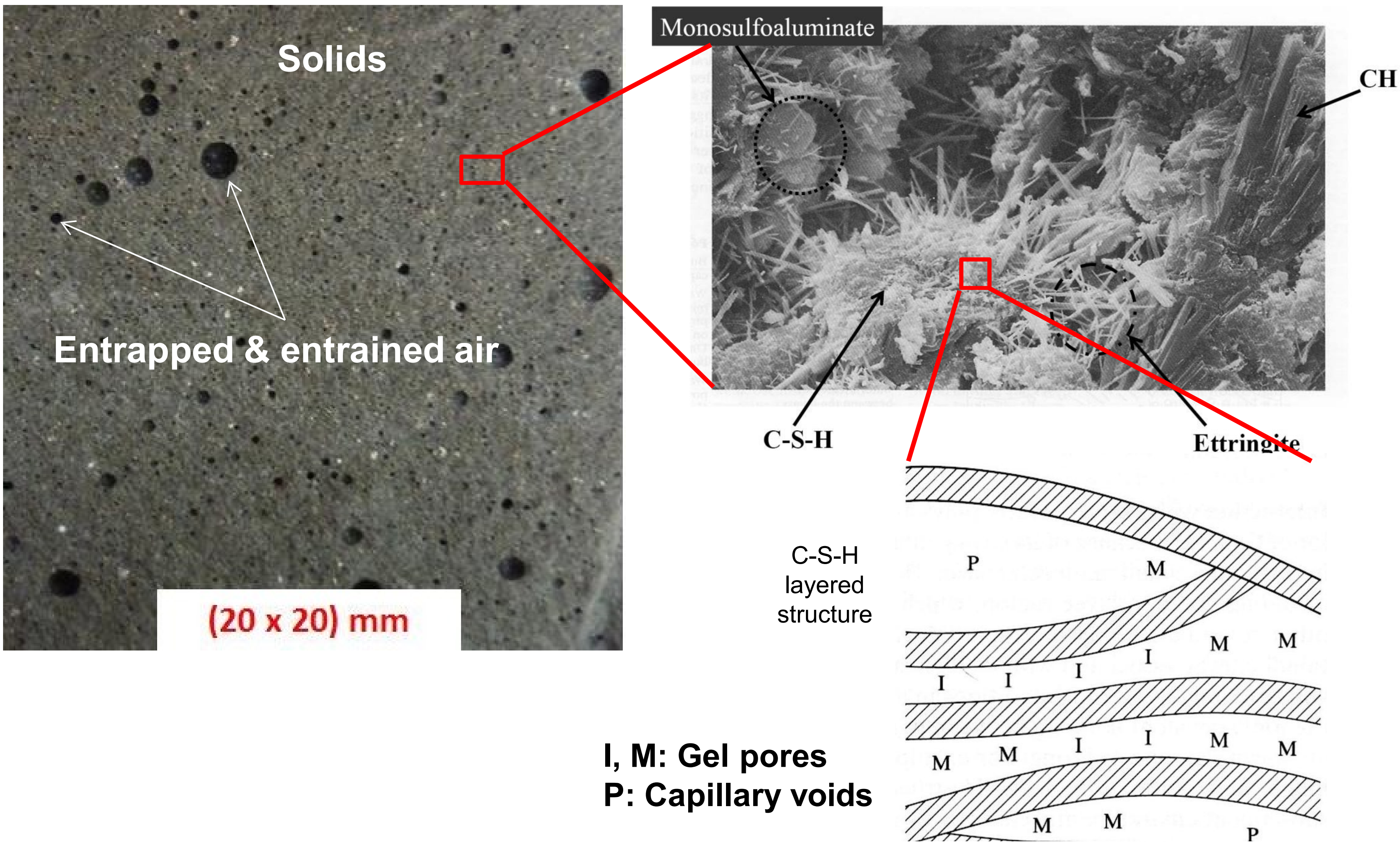
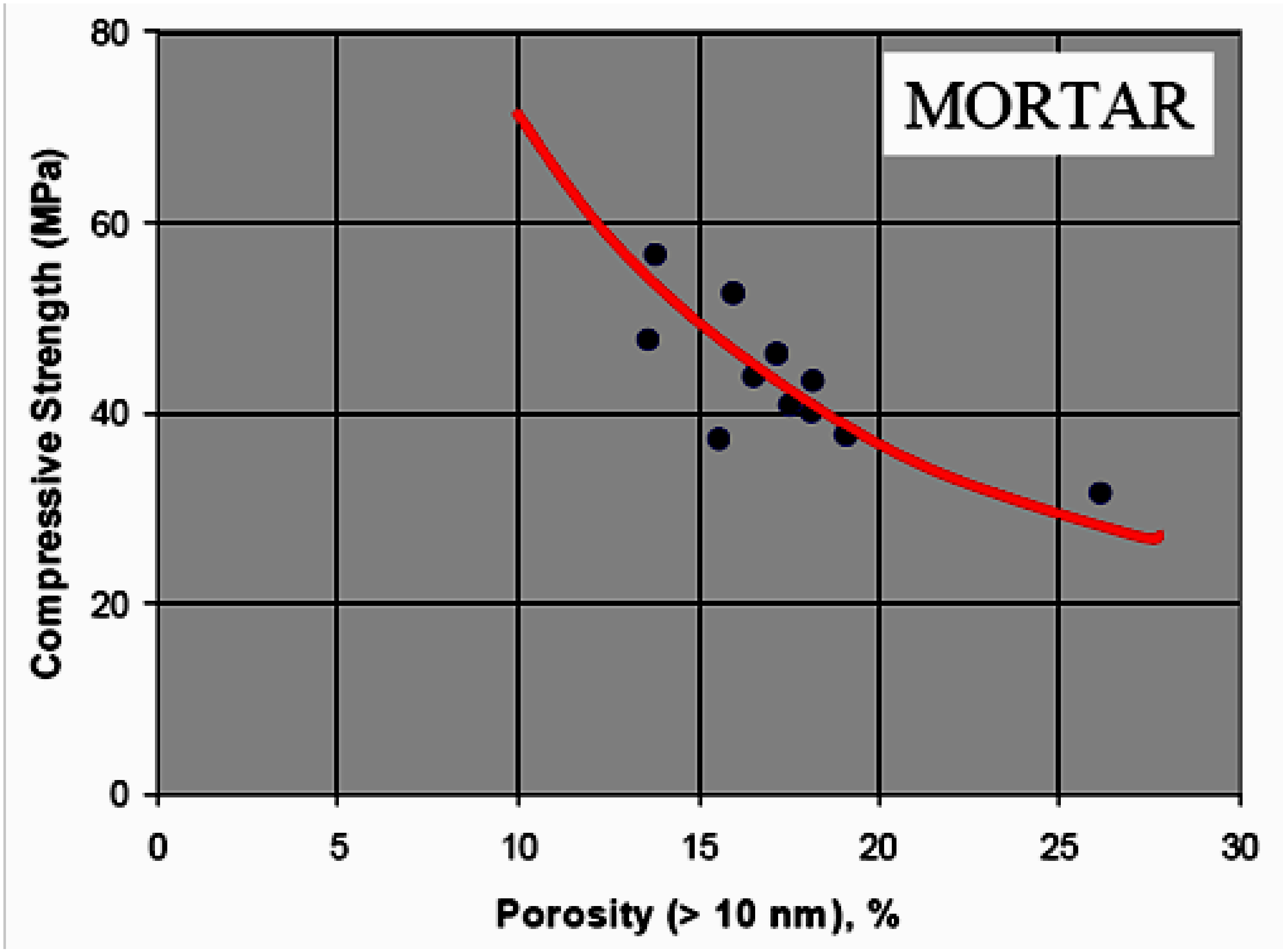
Structure of Hydrated Cement Paste



Voids in Hardened Cement Paste

- Entrapped air (> 1 mm)
  - Large pockets caused by handling
  - Decrease strength and increase permeability
- Entrained air (10 μm – 1 mm)
  - Microscopic bubbles caused by admixtures
  - No effect on permeability; improved durability
- Capillary voids (10 nm to 10 μm)
  - Extra water beyond hydration needs causes capillary voids
  - Reduces strength/stiffness and increases permeability
  - Depends on initial porosity (w/c ratio) and degree of hydration
- Gel pores (0.5 – 10 nm): Interlayer hydration space
  - Space between C-S-H gel atomic layers
  - Gel porosity = 28% (independent of w/c ratio and age)

Compressive Strength as A Function of Capillary Porosity



Water to Cement Ratio

- The most important property of hydrating cement

  Extra water beyond hydration needs causes capillary voids

  - Increases porosity and permeability
  - Decreases strength
  - Decreases durability

- Water to cement (W/C) ratio is expanded to water to cementitious material (W/CM) ratio or water to binder (W/B) ratio when supplementary cementitious materials are used as the binder system

Initial Porosity of Cement Paste

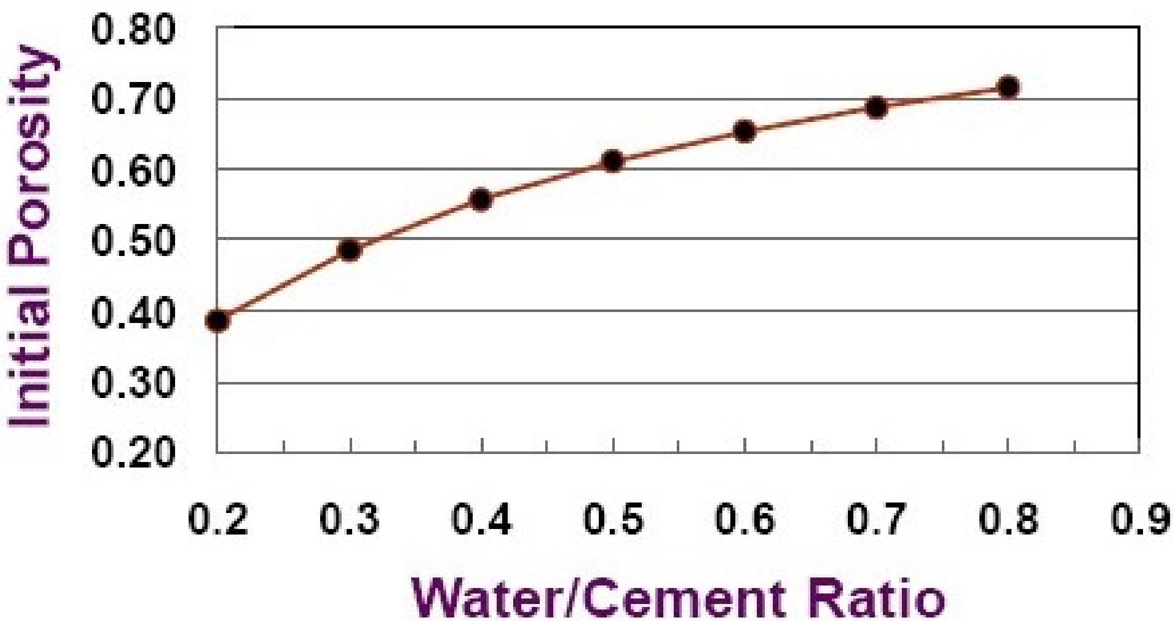
Initial Porosity of cement paste (p_i)

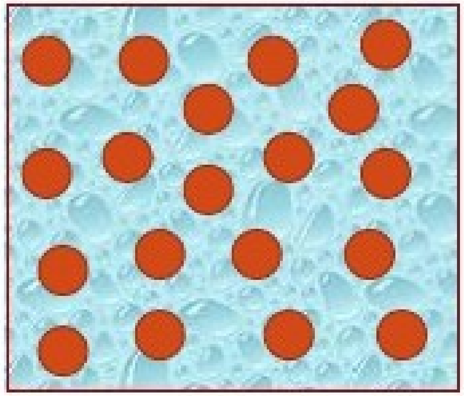
$= $ Volume of water / Volume of cement paste

$= (w \times C/1) / [(C / 3.15) + (w . C / 1)]$

$p_i = w / (w + 0.317)$

where w = W/C ratio; C = weight of cement;
Sp. gr. of cement = 3.15

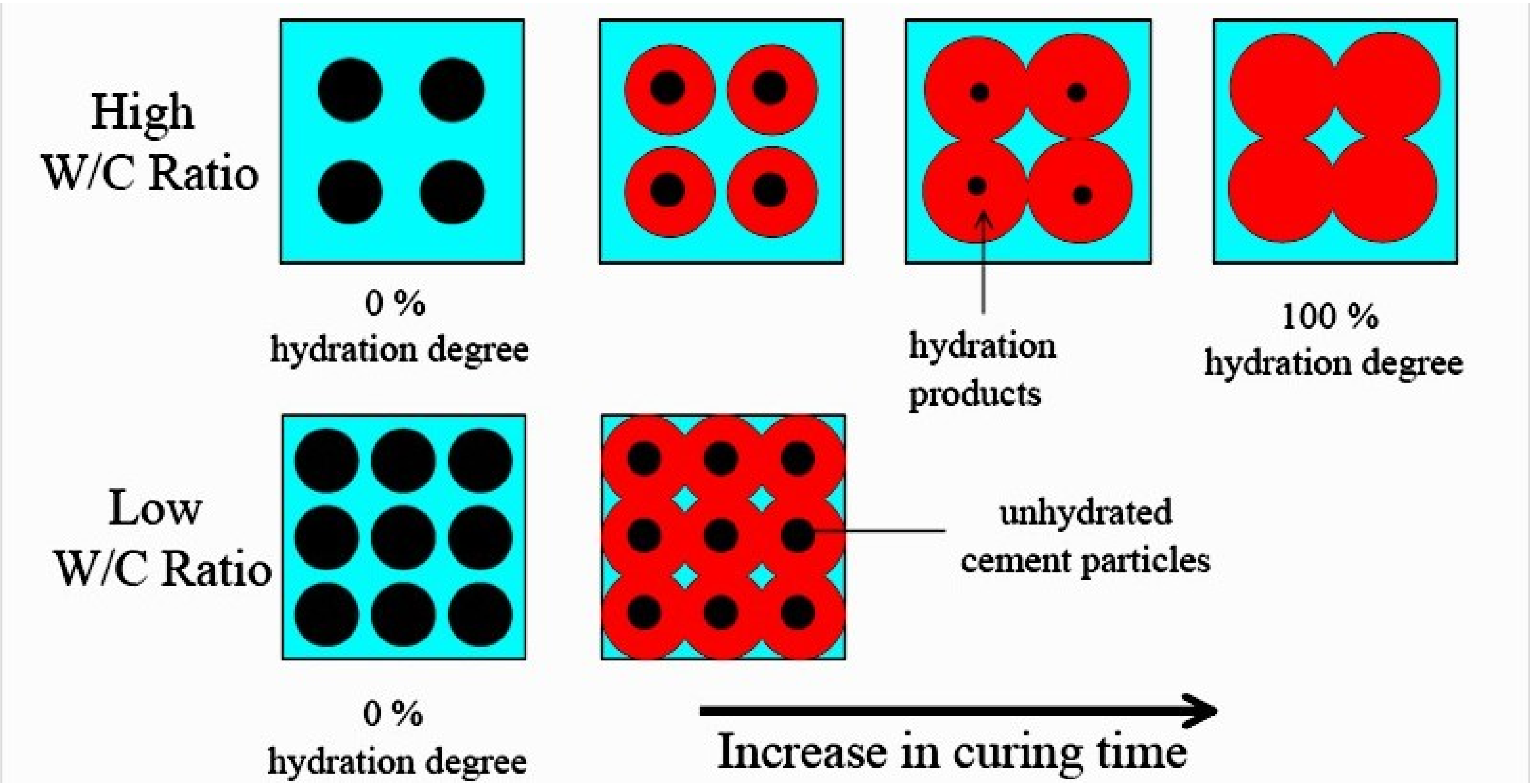




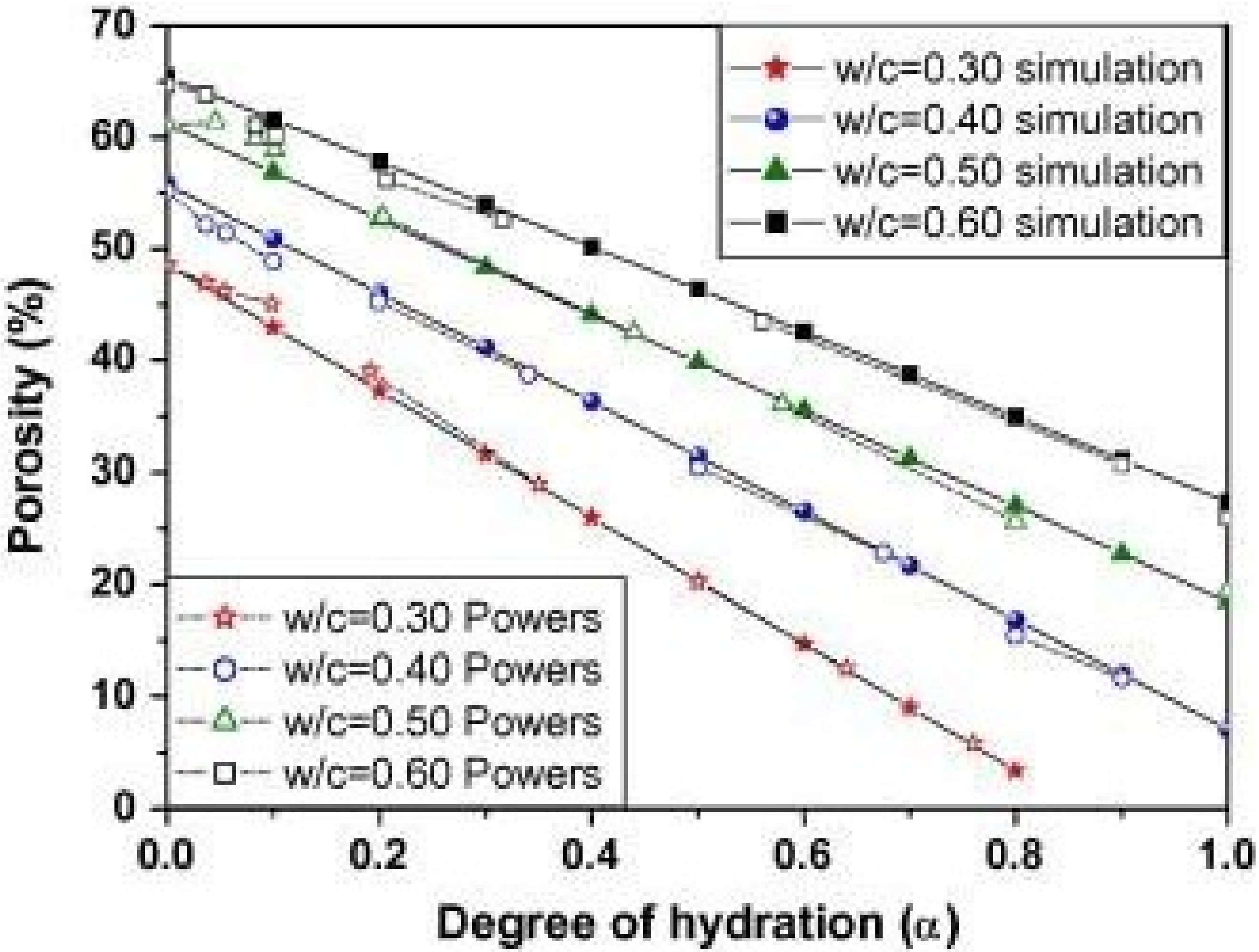
Cement particles in water

Degree of Hydration (Maturity)

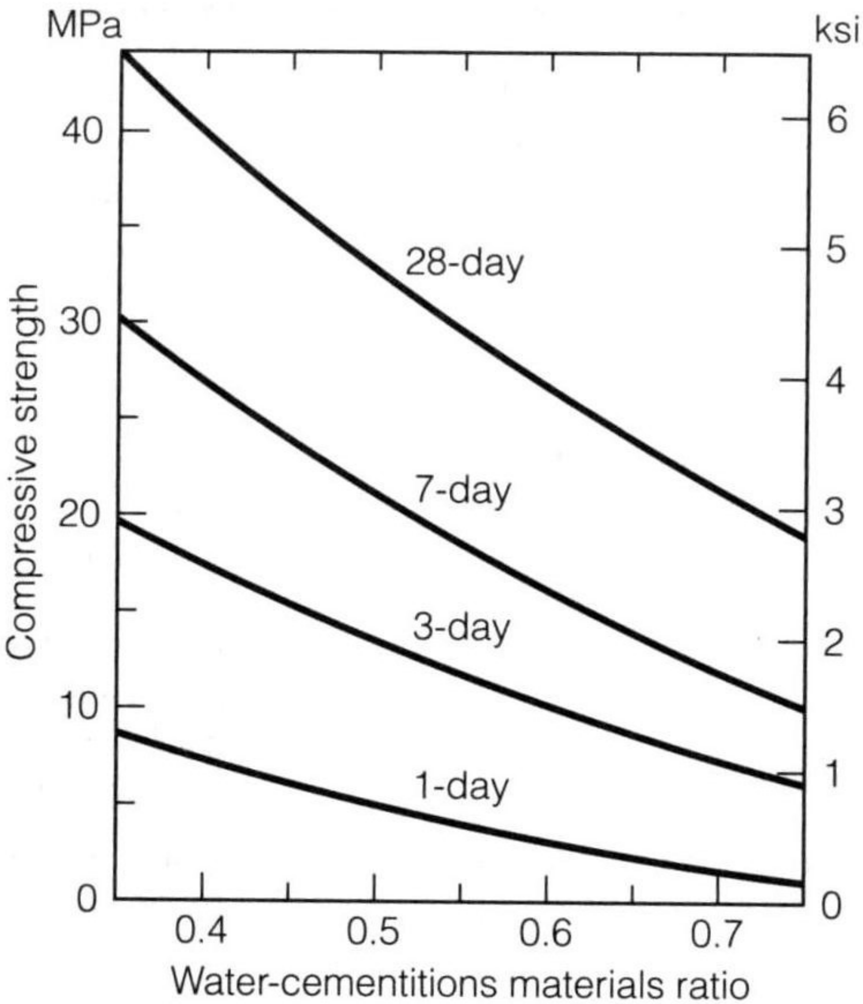
- Fraction of cement hydrated, α, ranging from 0 to 1



Capillary Porosity as A Function of Maturity



Abram's Law on Maturity and Concrete Compressive Strength



- Abram's law

$$\sigma = \frac{A}{B^{w/c}}$$